

RECRUITAI

PROJECT SPECIFICATION & IMPLEMENTATION BLUEPRINT

Fullstack Project Management Dashboard

Detailed project brief generated from the project record.

CANDIDATE

Varun Chopra

DURATION

30 days

TECHNOLOGY STACK

TypeScript

React

MySQL

REST API

Spring Boot

Project Overview

Summary of the problem, purpose and expected outcome.

PROJECT SUMMARY

A web based dashboard where teams can create projects, assign tasks, track progress, and view simple reports. The candidate will build a TypeScript React frontend, a Java Spring Boot REST API, and a MySQL database.

CANDIDATE Varun Chopra	TYPE fullstack	DURATION 30 days
--	--------------------------------------	--

Primary Objectives

- Implement a responsive React UI with TypeScript.
- Design and expose secure REST endpoints using Spring Boot.
- Model relational data in MySQL and enforce integrity.
- Add JWT based authentication and role based access control.
- Write unit, integration, and end to end tests.
- Containerize the app and deploy to a cloud VM.

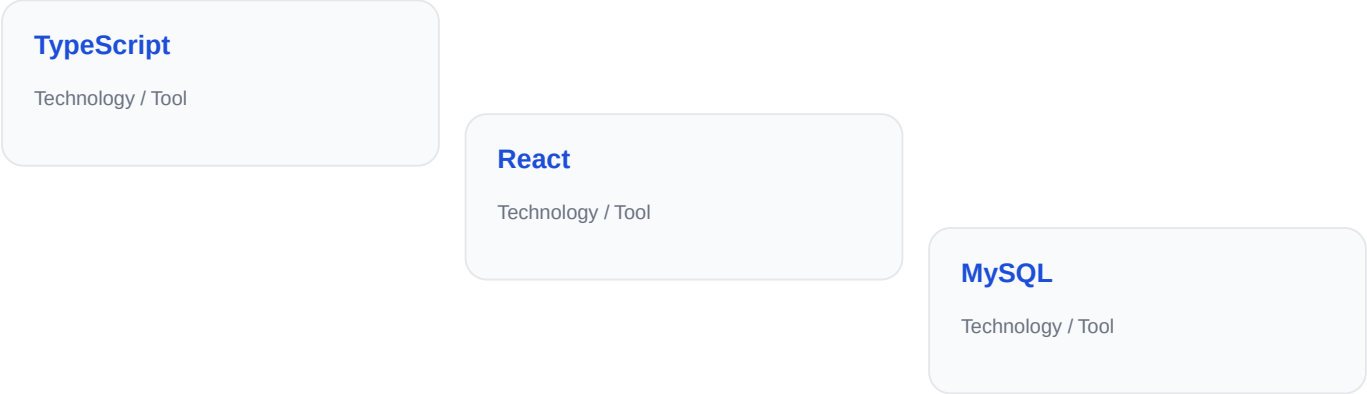
Candidate Details

NAME

Varun Chopra

Technology Stack

Technologies, frameworks and tools expected for implementation.



REST API

Technology / Tool

Spring Boot

Technology / Tool

Functional Requirements

- Register and login with email/password (JWT).
- Create, read, update, delete (CRUD) projects owned by a user.
- Within a project, CRUD tasks with status (To Do, In Progress, Done).
- Add comments to tasks; comments are visible to project members.
- Dashboard shows total tasks, completed tasks, and upcoming deadlines per project.
- Export project summary as CSV.

Non-Functional Requirements

- Passwords stored with bcrypt hashing.
- All API endpoints validate input and return proper HTTP status codes.
- Frontend must be responsive down to 320 px width.
- API response time < 200 ms for core endpoints under typical load.
- Automated tests must achieve "e80 % code coverage.

Modules & Features

Major functional areas that make up the project.

Authentication

Handles user sign up, login, JWT issuance, and role checks.

- User registration
- Password hashing
- Login endpoint
- JWT middleware
- Role based guard

Project Management

CRUD operations for projects and assignment of team members.

- Create project
- Edit project details
- Delete project
- List user projects
- Add/remove members

Task Board

Manage tasks within a project, track status, and add comments.

- Create task

- Update task status
- Delete task
- Add comment
- List task comments

Reporting

Generate simple analytics and CSV export for a project.

- Project summary endpoint
- CSV export
- Progress chart data

Frontend UI

React components for all user interactions and visualizations.

- Auth pages
- Project list view
- Kanban board
- Comment thread
- Dashboard charts
- CSV download

Implementation Roadmap

Recommended phases and milestones for completing the project.

1

Foundation

Initialize Git monorepo, Create React TS app, Create Spring Boot project, Add Dockerfiles, Configure GitHub Actions

- Initialize Git monorepo
- Create React TS app
- Create Spring Boot project
- Add Dockerfiles
- Configure GitHub Actions

2

Authentication

Design User entity, Implement register/login endpoints, Add bcrypt hashing, Create auth React pages, Integrate JWT storage

- Design User entity
- Implement register/login endpoints
- Add bcrypt hashing
- Create auth React pages
- Integrate JWT storage

3

Core APIs

Design Project, Task, Comment tables, Implement CRUD controllers, Add validation, Write service layer tests

- Design Project, Task, Comment tables
- Implement CRUD controllers
- Add validation
- Write service layer tests

4

Frontend Integration

Create project list component, Build Kanban board, Implement comment thread, Add CSV export button, Handle auth token

- Create project list component
- Build Kanban board
- Implement comment thread
- Add CSV export button
- Handle auth token

5

Reporting & Export

Add summary endpoint, Generate CSV stream, Create dashboard charts, Write integration tests

- Add summary endpoint
- Generate CSV stream
- Create dashboard charts
- Write integration tests

6

Testing & Deployment

Write E2E Cypress tests, Add rate limiting filter, Configure Docker Compose for prod, Deploy to cloud VM

- Write E2E Cypress tests
- Add rate limiting filter
- Configure Docker Compose for prod
- Deploy to cloud VM

Evaluation Criteria

Criteria that can be used to assess the completed project.

- Criterion: Functional Completeness | Description: All listed functional requirements are implemented and work as described | Weight: 30
- Criterion: Code Quality | Description: Clean, modular code with proper TypeScript typings and Java conventions | Weight: 15
- Criterion: Security Implementation | Description: Password hashing, JWT handling, input validation, and role checks | Weight: 15

- Criterion: Testing Coverage | Description: Unit, integration, and E2E tests reaching "e80 % coverage | Weight: 15
- Criterion: Performance & Responsiveness | Description: API latency <200 ms; UI remains responsive on mobile | Weight: 10
- Criterion: Documentation & Deployability | Description: Clear README, API docs, DB schema, and deployment scripts | Weight: 15

Expected Deliverables

Artifacts and outcomes expected at project completion.

- Source code repository (frontend & backend)
- README with setup and run instructions
- OpenAPI/Swagger documentation
- MySQL schema diagram
- Automated test suite
- Docker Compose file for local and production deployment

Candidate Alignment

Project requirements compared with the candidate information available in the project record.

Varun already knows TypeScript, Angular, Java, and Spring Boot. Switching the frontend to React leverages his TypeScript expertise while exposing him to a new UI library. His Java/Spring background fits the REST API layer. The 30 day scope matches a mid level challenge: it requires fullstack design, JWT auth, relational modeling, testing, and containerized deployment without overwhelming complexity.

Implementation Note

This document is generated from the project record. AI-generated requirements should be reviewed by an administrator before implementation.